

Kihyeon Joo

M.S./Ph.D. Integrated Student · Department of Electrical and Computer Engineering, Inha University

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RESEARCH INTERESTS

Robotics (vision-language-action models, human preference learning) · Human-Robot Interaction · Ethical AI (educational robots for moral development) · Brain-Computer Interfaces (EEG emotion recognition, EEG foundation models)

EDUCATION

Inha University, Incheon, South Korea Mar. 2024 – Present
M.S./Ph.D. Integrated Program in Electrical and Computer Engineering · Advisor: Prof. Byung Hyung Kim

Inha University, Incheon, South Korea 2024
B.S. in Information and Communication Engineering · Advisor: Prof. Yongwoo Lee

PUBLICATIONS

Domestic Journals (in Korean)

- [J2] **K. Joo**, C. Woo, and B. H. Kim, “Diffusion-Based Data Augmentation for Few-Channel EEG Emotion Recognition Using Class-Conditional DDPM,” *Journal of Biomedical Engineering Research*, vol. 47, no. 1, pp. 1–10, Feb. 2026. [\[link\]](#)
- [J1] **K. Joo** and B. H. Kim, “AEGIS: A mmWave-based Wearable System for Efficient Human Detection in Low-Visibility Rescue Environments,” *Journal of Korea Multimedia Society*, vol. 28, no. 12, pp. 1914–1929, Dec. 2025. [\[doi\]](#)

Domestic Conferences (in Korean)

- [C1] J. Yuk, **K. Joo**, Y. Park, and B. H. Kim, “Contact-Pressure-Invariant PPG Signal Extraction Model,” in *Proc. Korea Computer Congress (KCC)*, pp. 1151–1153, Jun. 2023.

RESEARCH EXPERIENCE

Graduate Researcher, Affective Artificial Intelligence Lab, Inha University Mar. 2024 – Present

- **Robotics**: Vision-language-action models for robotic manipulation on a UR5e arm — a visual contextual reasoning framework and human preference learning.
- **Human-Robot Interaction**: How a robot's speaking style affects continuous collaboration and whether people choose to work with it again.
- **Ethical AI**: ROBI, an educational robot for children's moral development.
- **Brain-Computer Interfaces**: Few-channel EEG data augmentation [J2] and calibration-aware fine-tuning of EEG foundation models.
- **Wearable Sensing**: mmWave-based wearable system for human detection in low-visibility rescue environments [J1].

RESEARCH PROJECTS

Development of XR Content Agent Technology Based on Emotion and Affect Reasoning Apr. 2026 – Dec. 2028
Institute of Information & Communications Technology Planning & Evaluation (IITP) · Lead Researcher

HONORS & SCHOLARSHIPS

Inha Vision Fellowship, Inha University	2026 – Present
AI Convergence Innovation Human Resources Development Scholarship, IITP / MSIT	2024 – 2025

TEACHING EXPERIENCE

Teaching Assistant, Reinforcement Learning, Inha University	2024
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SKILLS

Programming	Python, PyTorch, Hugging Face
Robotics	ROS2, MoveIt2, Isaac Sim, Intel RealSense, UR5e
BCI / EEG	MNE-Python, Braindecode, scikit-learn
Environment	Ubuntu, CUDA, Conda

Last updated: October 2026